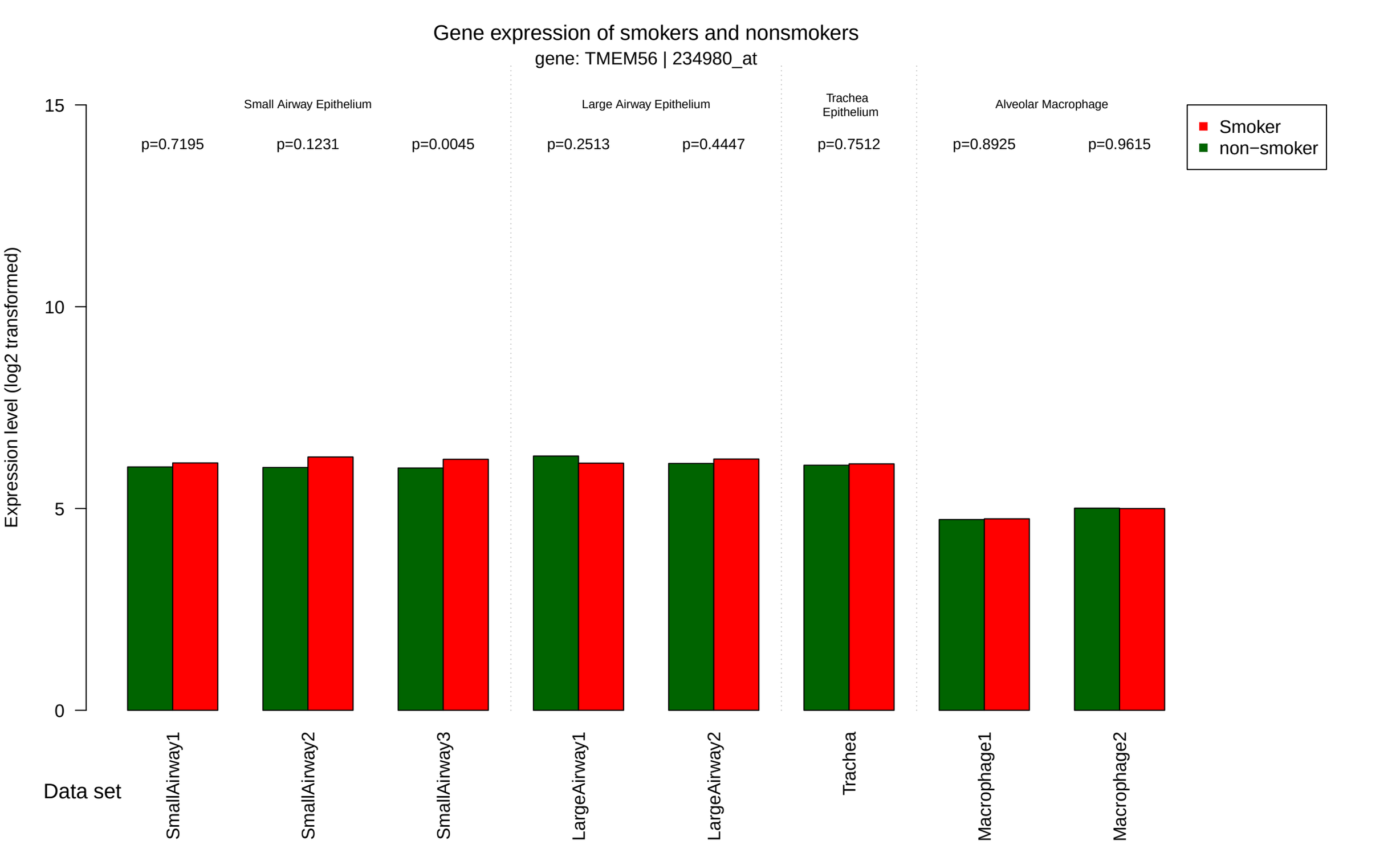
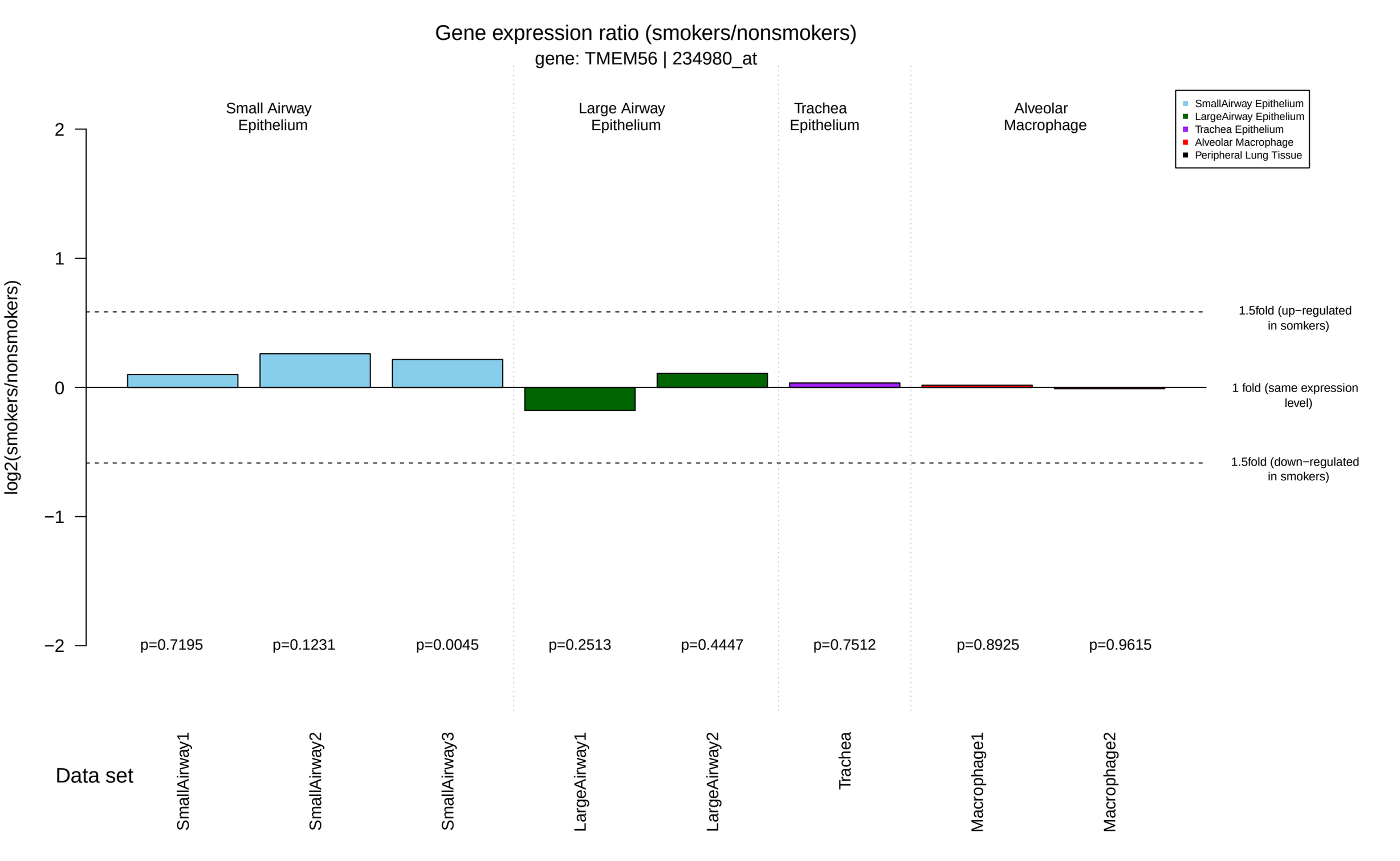
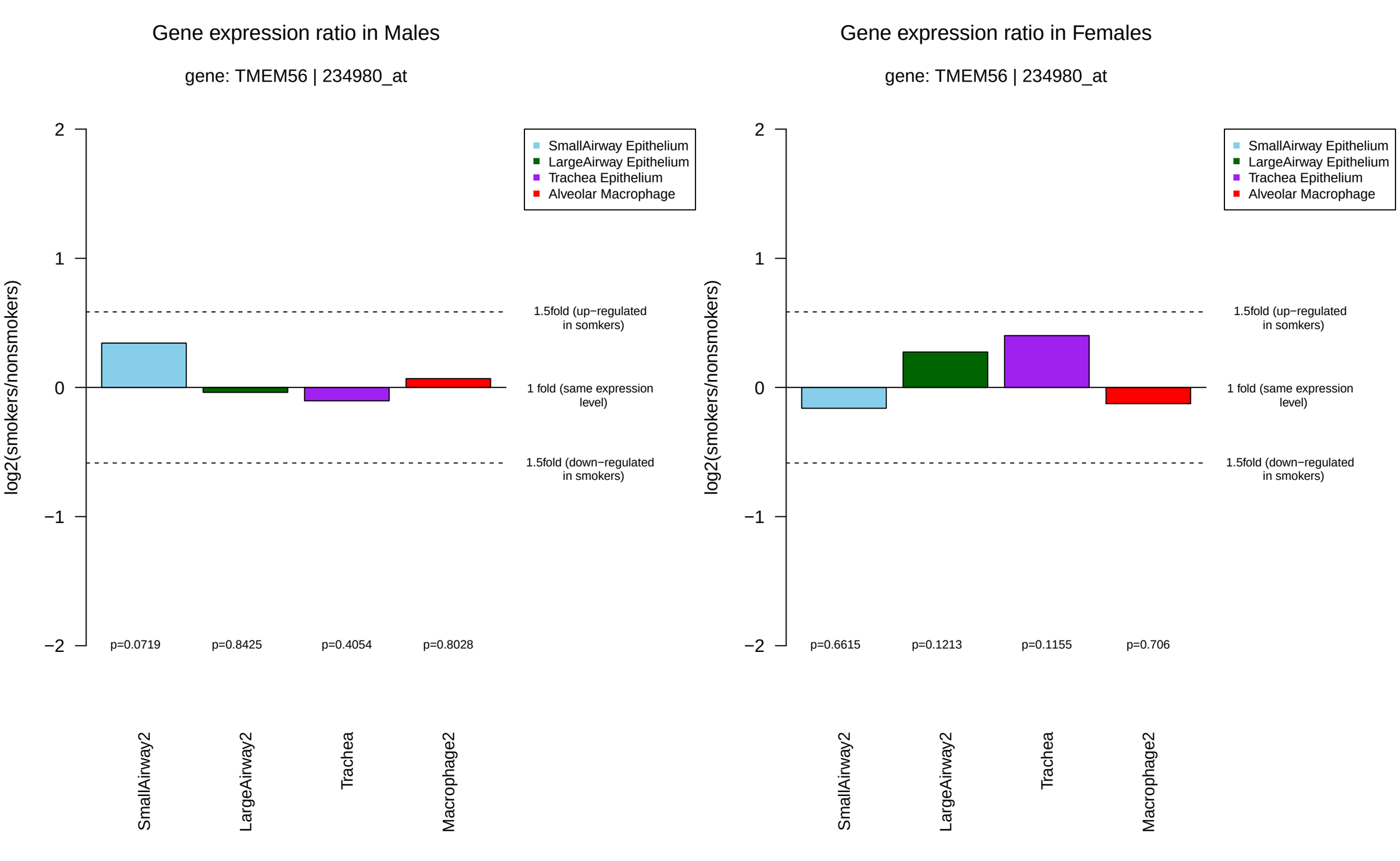
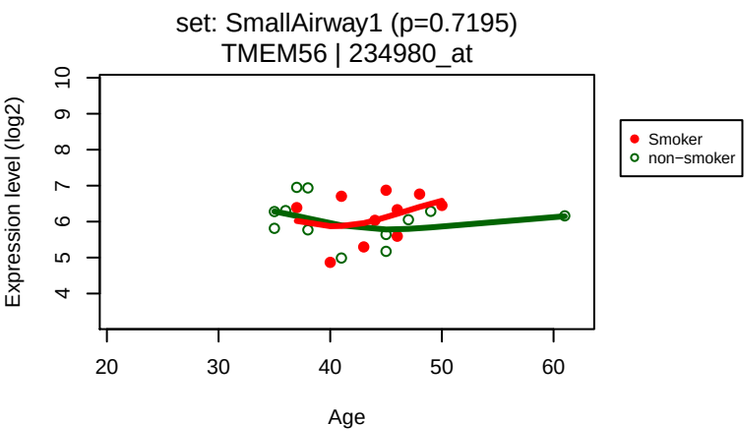
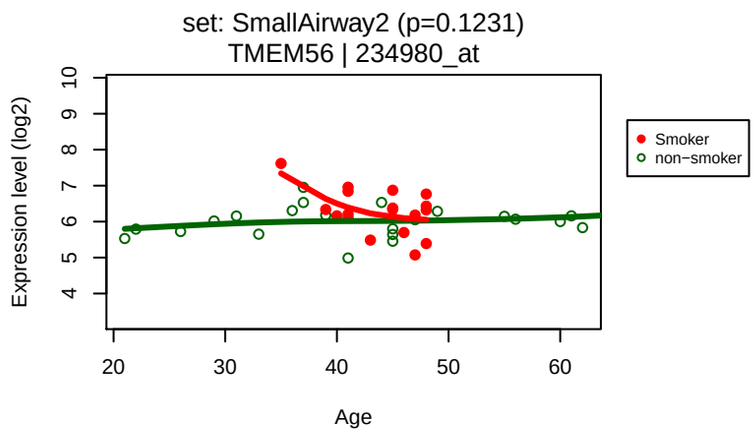
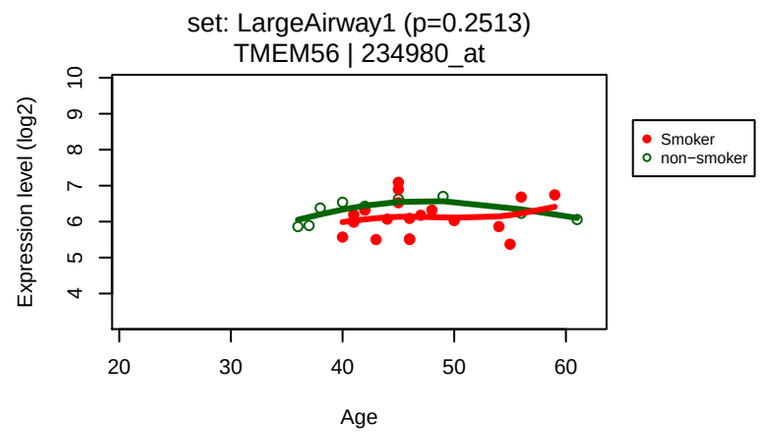

Gene expression of smokers and nonsmokers
gene: TMEM56 | 234980_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7195
p=0.1231
p=0.0045
p=0.2513
p=0.4447
p=0.7512
p=0.8925
p=0.9615
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

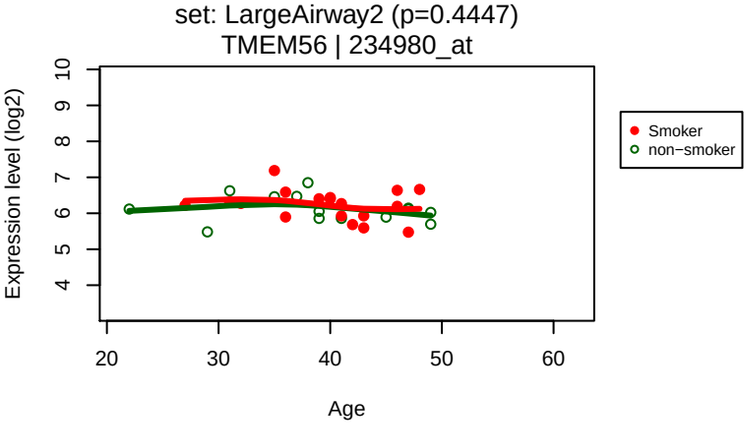

Gene expression ratio (smokers/nonsmokers)
gene: TMEM56 | 234980_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7195
p=0.1231
p=0.0045
p=0.2513
p=0.4447
p=0.7512
p=0.8925
p=0.9615
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

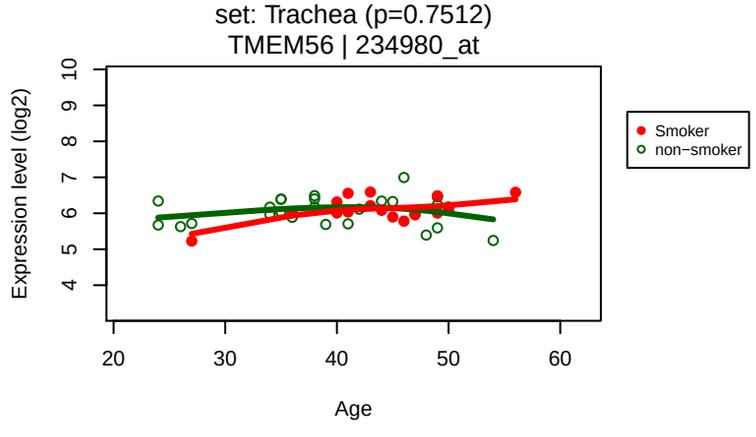

Gene expression ratio in Males
gene: TMEM56 | 234980_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0719
p=0.8425
p=0.4054
p=0.8028
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM56 | 234980_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6615
p=0.1213
p=0.1155
p=0.706
SmallAirway2
LargeAirway2
Trachea
Macrophage2

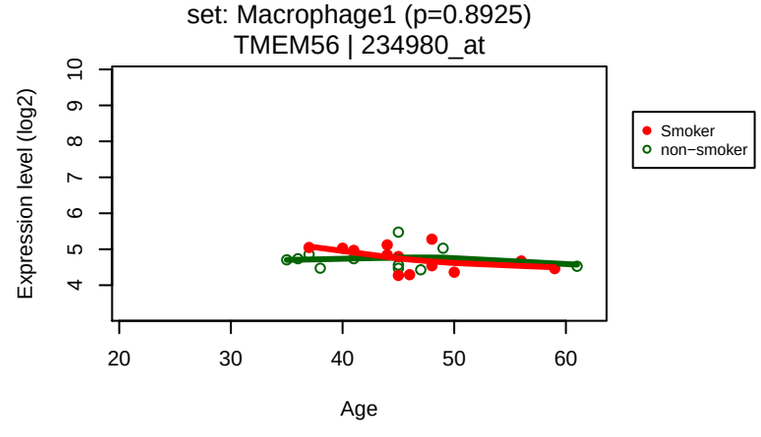

set: SmallAirway1 (p=0.7195)
TMEM56 | 234980_at
Expression level (log2)
Age
Smoker
non−smoker

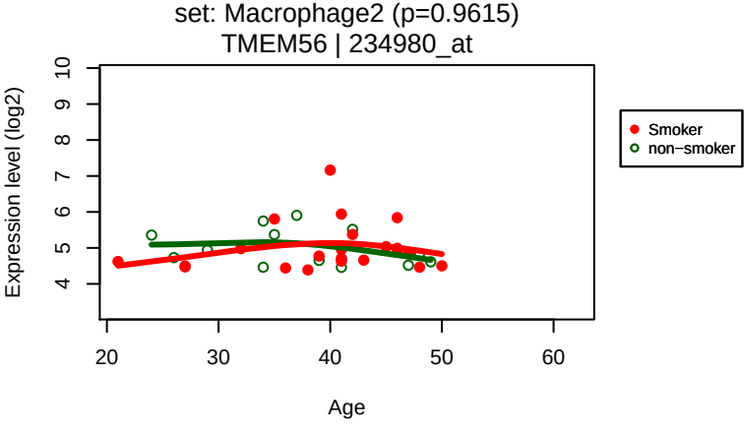

set: SmallAirway2 (p=0.1231)
TMEM56 | 234980_at
Expression level (log2)
Age
Smoker
non−smoker

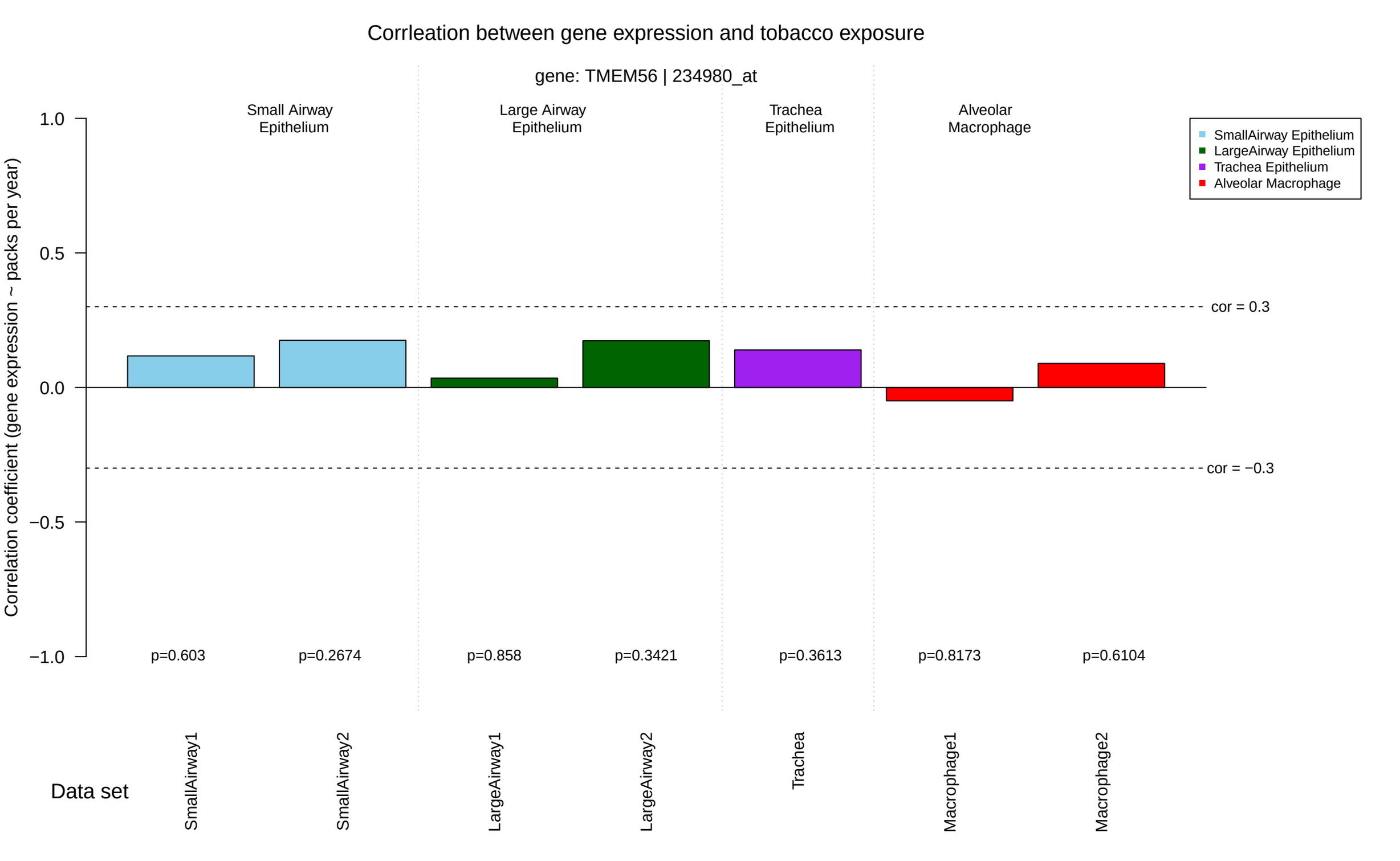

set: LargeAirway1 (p=0.2513)
TMEM56 | 234980_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4447)
TMEM56 | 234980_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7512)
TMEM56 | 234980_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8925)
TMEM56 | 234980_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9615)
TMEM56 | 234980_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM56 | 234980_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.603
p=0.2674
p=0.858
p=0.3421
p=0.3613
p=0.8173
p=0.6104
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2